

Lab 0 - Rubber Stoppers

Aaron Huang

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1 Purpose

1. To determine the density of two sizes of rubber stoppers
2. To write up a complete lab report using proper format

2 Introduction

Density was defined as a physical property that described the relationship between the mass of a material and its volume, calculated using $\rho = \frac{m}{V}$. In this formula the density was represented by ρ , the mass was represented by m , and the volume was represented by V .

Water-displacement and mass measurements were used to determine the volumes and densities of two rubber stoppers. The smaller stopper volume was determined by graduated-cylinder displacement; the larger stopper volume was determined by measuring the change in mass of a watch glass after overflow collection. Calculations were performed using measured masses and displaced-volume values to obtain densities.

There were no special safety considerations for this experiment.

3 Materials

- Graduated cylinder (200 mL)
- Two rubber stoppers (small & large)
- Beaker (50 mL)
- Watch glass
- Water
- Mass scale
- Dropper

3.1 Apparatus

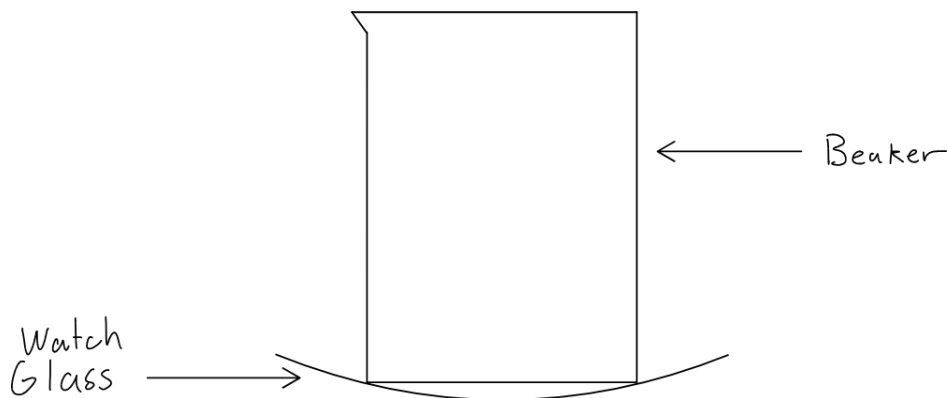


Figure 1: Apparatus used for part two of the experiment

4 Procedure

<u>Materials/Equipment</u>	<u>Actions Taken</u>
Mass Scale	Mass of the large rubber stopper was measured and recorded Initial mass of the watch glass was measured and recorded
Beaker (50 mL)	Beaker was filled leaving handling clearance Beaker was placed on the watch glass to collect overflow
Dropper	Water was added using a dropper until beaker was full
Large Stopper	Stopper was slowly and carefully lowered into the beaker
Dropper	Excess water was removed to avoid spillage
Mass Scale	Final mass of the watch glass with collected water was measured and recorded

Table 1: Observations of Rubber Stoppers — Small

Mass (g)	Initial Measurement (mL)	Final Measurement (mL)
4.33	15.1	16.8

5 Observations

Table 2: Observations of Rubber Stoppers — Large

Mass (g)	Initial Measurement (g)	Final Measurement (g)
24.25	25.59	41.07

6 Analysis

$$\begin{aligned}
 \text{Volume} &= \text{Final} - \text{Initial} \\
 &= 16.8 \text{ mL} - 15.1 \text{ mL} \\
 &= 1.7 \text{ mL} \\
 &= 1.7 \text{ cm}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Density} &= \frac{\text{Mass}}{\text{Volume}} \\
 &= \frac{4.33 \text{ g}}{1.7 \text{ cm}^3} \\
 &\approx 2.6 \text{ g/cm}^3
 \end{aligned}$$

The density of water was assumed to be 1 g/mL, the calculation for the large stopper was similar.

Table 3: Summary of Density Results

Rubber Stopper	Density (g/cm ³)
Small	2.6
Large	1.567

7 Discussion

1. With data in Table 3, the percentage error for the method from Part 2 was calculated as follows:

$$\begin{aligned}
 \text{Percentage Error} &= \left| \frac{\text{Experimental Value} - \text{Theoretical Value}}{\text{Theoretical Value}} \right| \times 100\% \\
 &= \left| \frac{1.567 - 2.6}{2.6} \right| \times 100\% \\
 &\approx 40\%
 \end{aligned}$$

$\therefore$ the percentage error for the method from Part 2 was approximately 40%.

The error in the method from Part 2 could be attributed to several factors. One significant source of error was the surface tension of the water, which may have caused the water to cling the bottom of the beaker when it was removed from the watch glass, leading to an underestimation of the volume of water displaced by the large rubber stopper.

2. To find the percentage composition of copper and zinc in a brass sample, a similar procedure to the one used for the rubber stoppers may be followed. To begin, the mass of the brass sample would be taken using a mass scale. Then the volume of the brass sample would be determined using water displacement in a graduated cylinder. Next, the density would be calculated using the formula $\rho = \frac{m}{V}$. Lastly, using the known densities of pure copper and pure zinc, a system of equations would be used to solve for the percentage composition of copper and zinc in the brass sample.

8 Conclusion

The density of the small rubber stopper was approximately 2.6 g/cm^3 , and the density of the large rubber stopper was approximately 1.567 g/cm^3 , with a percentage error of 40%. A lab report was written with proper formatting.